AI-STUDENT-INSIGHTS

Exploratory Data Analysis Report

An exploratory study of generative AI usage, academic performance, study habits, exam anxiety, and student well-being.

Report generated: 28 October 2026

Dataset records: 50,000

Dataset variables: 16

Prepared as a data analysis project using Python and exploratory statistical techniques.

1. Abstract

This project explores a dataset concerning generative artificial intelligence usage and student-related outcomes. The analysis examines usage patterns, academic performance, traditional study habits, perceived AI dependency, exam anxiety, skill retention, and burnout risk. The workflow includes dataset inspection, data-quality checks, descriptive statistics, distribution analysis, correlation analysis, and group-wise comparisons. Visualizations are used to identify patterns and relationships that may warrant further investigation. The findings are descriptive and do not establish that AI usage causes any particular academic or well-being outcome.

2. Introduction

Generative AI tools are increasingly used for tasks such as debugging, drafting, summarizing, brainstorming, and answering questions. Understanding how these usage patterns relate to student outcomes can help identify trends worth investigating. Exploratory Data Analysis (EDA) provides a systematic way to inspect data, summarize variables, visualize distributions, and investigate relationships before building predictive models.

3. Problem Statement

The dataset contains multiple measures of AI usage, academic performance, study behavior, and student well-being. Without systematic analysis, it is difficult to understand how these variables are distributed and whether observable associations exist between them. This project addresses that need through structured exploratory analysis and data visualization.

4. Objectives

- Inspect the structure and quality of the processed dataset.
- Summarize numerical variables using descriptive statistics.
- Examine categorical distributions and AI usage groups.
- Compare pre-semester and post-semester GPA values.
- Investigate relationships among AI usage, dependency, anxiety, study hours, and academic outcomes.
- Explore differences in skill retention and burnout risk across groups.
- Communicate findings through clear visualizations.

5. Research Questions

- How are students distributed across AI usage groups?
- How do pre-semester and post-semester GPA values compare?
- Is weekly AI usage associated with perceived AI dependency?
- How does exam anxiety vary across AI usage groups?
- How do skill-retention scores differ across AI usage groups?
- How are AI usage hours distributed across burnout-risk categories?

6. Dataset Overview

The processed dataset contains 50,000 records and 16 columns. The analysis uses the processed CSV provided in the project's data/processed directory.

Dataset property	Result
Number of records	50,000
Number of columns	16
Missing cells	0
Duplicate rows	0
Student_ID column	Not present
Unused AI_Use_Group column	Not present

Variables Included

- Major Category
- Year of Study
- Pre Semester GPA
- Weekly GenAI Hours
- Primary Use Case
- Prompt Engineering Skill
- Tool Diversity
- Paid Subscription
- Traditional Study Hours
- Perceived AI Dependency
- Institutional Policy

• Anxiety Level During Exams

• Post Semester GPA

• Skill Retention Score

• Burnout Risk Level

• AI Usage Group

7. Data Quality Assessment

- The script identified 0 empty or missing cells and 0 duplicate rows in the processed dataset. These checks are based on the CSV currently loaded by this script. They do not independently prove that every value is valid, correctly labeled, or representative.
- Additional checks should confirm numerical ranges, categorical labels, the derivation of AI_Usage_Group, and the provenance of the source data. The original Student_ID field should generally be excluded from statistical analysis when it functions only as an identifier.

8. Descriptive Statistics

Variable	N	Mean	Median	Std. dev.	Min	Max
Pre Semester GPA	50000	3.15	3.21	0.48	1.18	4.00
Weekly GenAI Hours	50000	8.43	5.80	8.27	0.00	40.00
Tool Diversity	50000	2.80	3.00	1.19	1.00	5.00
Traditional Study Hours	50000	11.21	11.18	5.16	1.00	35.86
Perceived AI Dependency	50000	3.51	3.00	1.82	1.00	10.00
Anxiety Level During Exams	50000	4.27	4.00	2.14	1.00	10.00
Post Semester GPA	50000	3.35	3.42	0.50	1.00	4.00
Skill Retention Score	50000	75.80	76.00	13.28	10.78	100.00

These statistics describe the values in the processed dataset. The standard deviation shown is the sample standard deviation.

9. Categorical Analysis

10. Academic Performance Analysis

- The mean pre-semester GPA is 3.15, while the mean post-semester GPA is 3.35.
- The mean paired GPA change is 0.20 points, calculated as post-semester GPA minus pre-semester GPA for rows with valid values in both columns.

GPA change category	Count	Percentage
Improved	43,759	87.52%
Unchanged	49	0.10%
Declined	6,192	12.38%

A change in GPA between two measurements does not establish that AI usage caused the change. The report also does not assume that the GPA scale or measurement process is valid without verification from the dataset documentation.

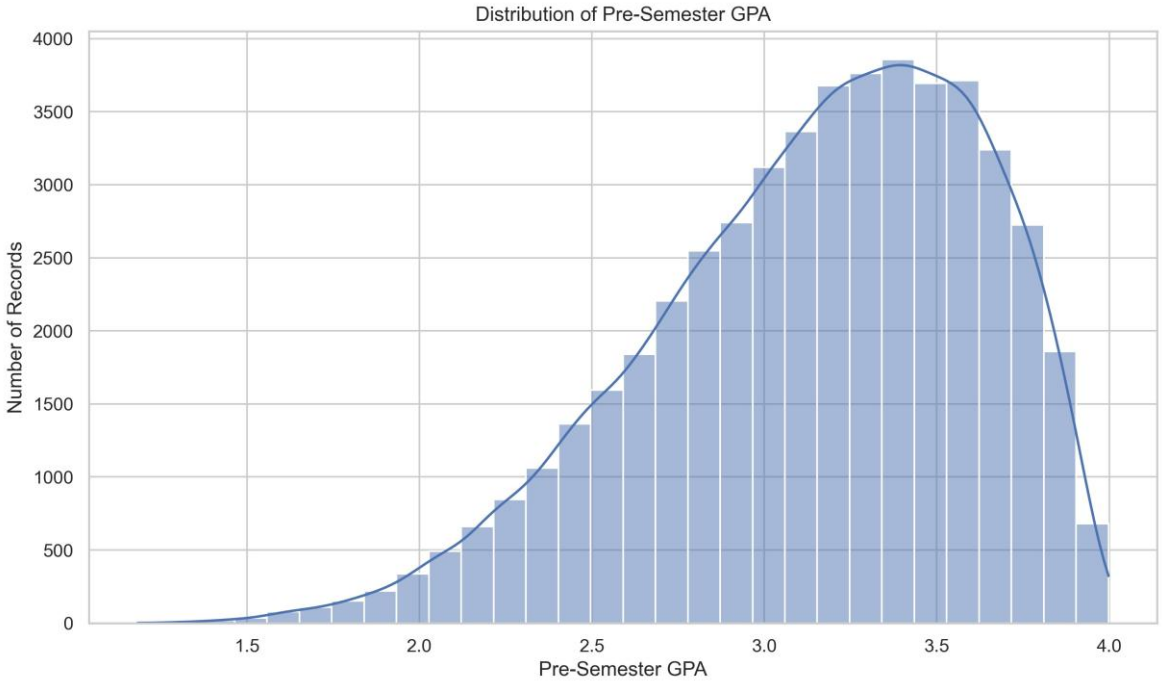


Figure 10.1: Distribution of pre-semester GPA.

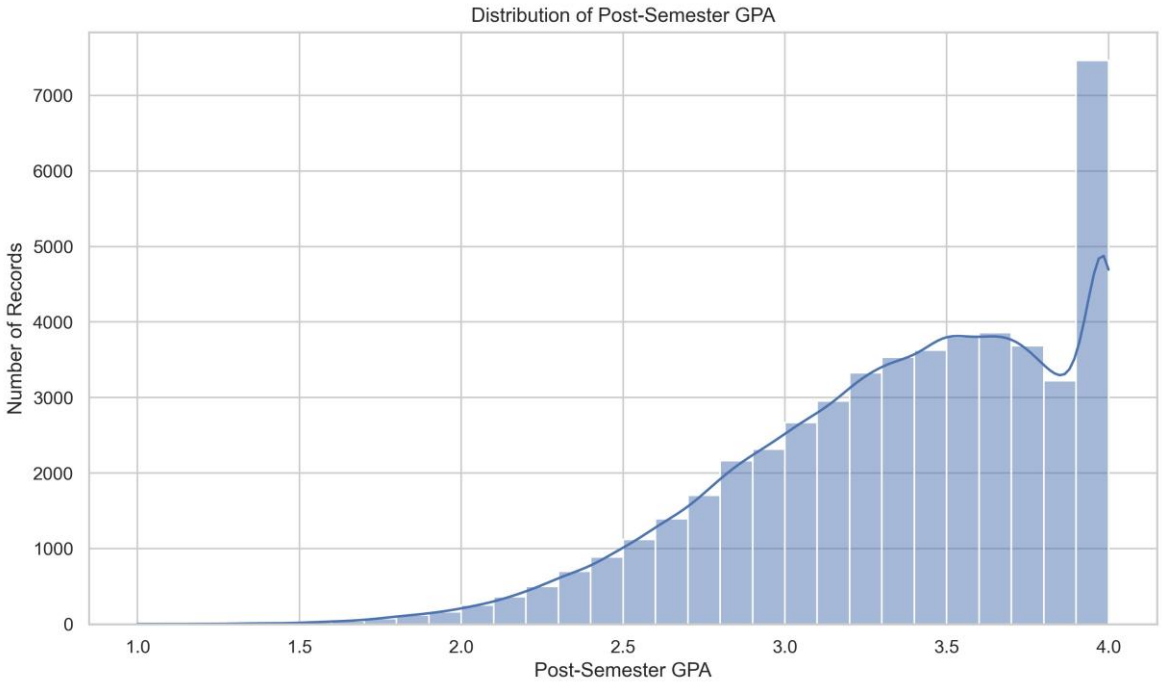


Figure 10.2: Distribution of post-semester GPA.

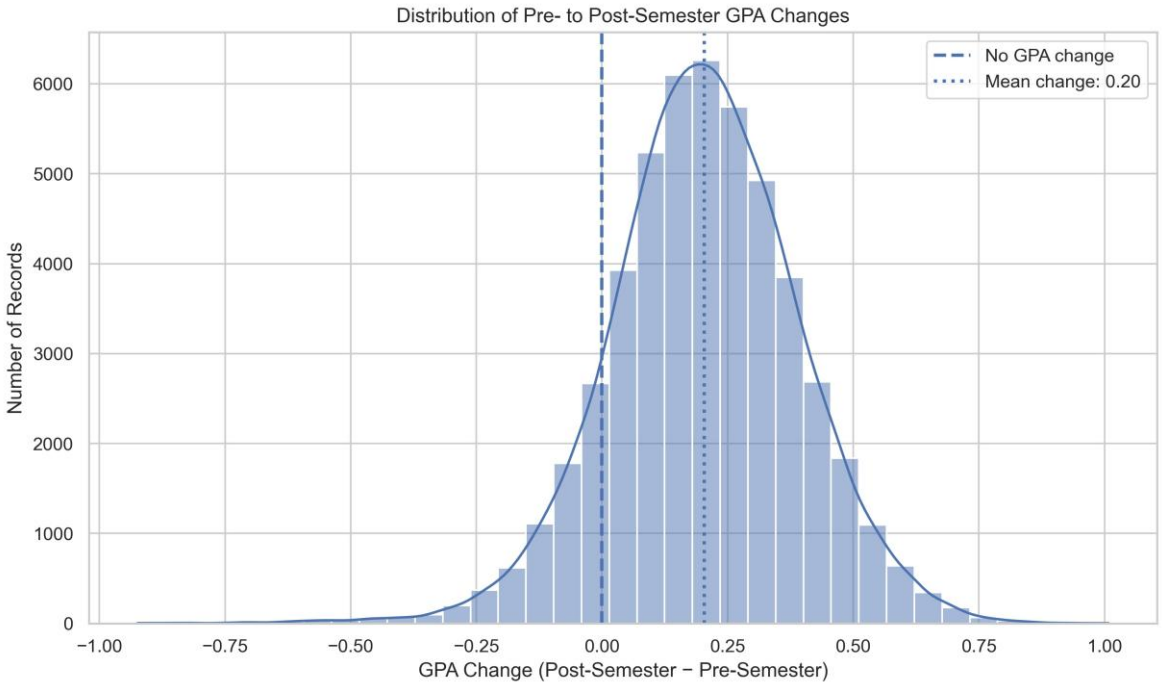


Figure 10.3: Distribution of changes between the two GPA measurements.

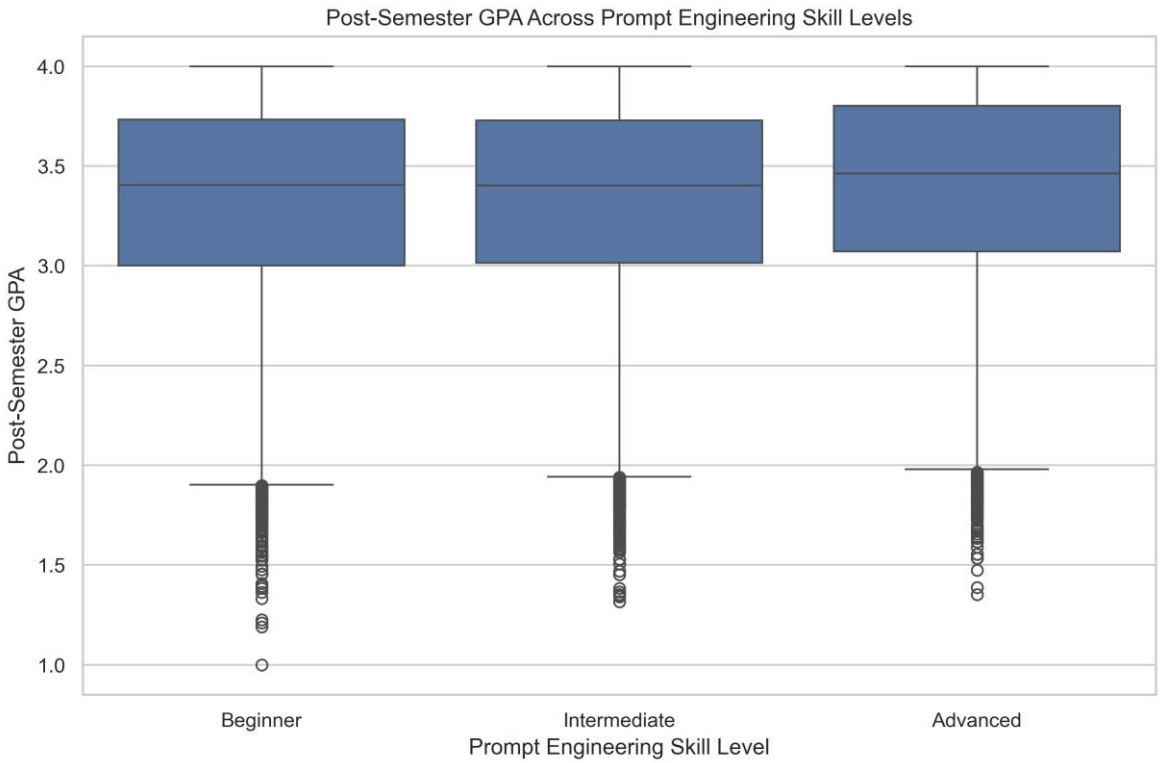


Figure 10.4: Post-semester GPA by prompt-engineering skill level.

11. AI Usage and Student Well-Being

This section examines AI usage alongside perceived AI dependency, exam anxiety, traditional study hours, and burnout-risk categories. Group comparisons describe differences in the observed data and should not be interpreted as causal effects.

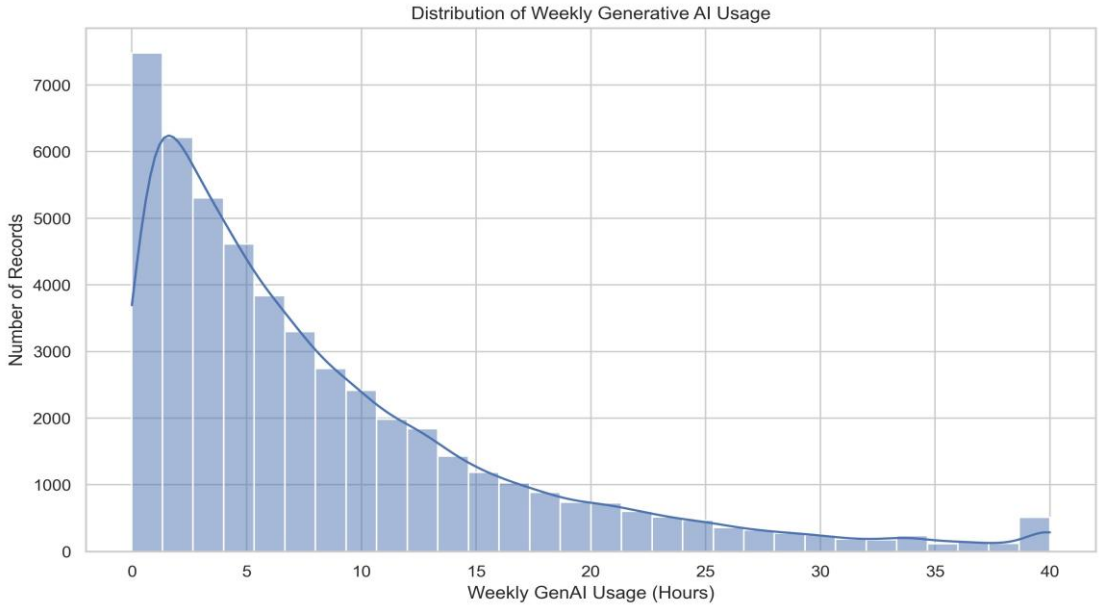


Figure 11.1: Distribution of weekly generative AI usage hours.

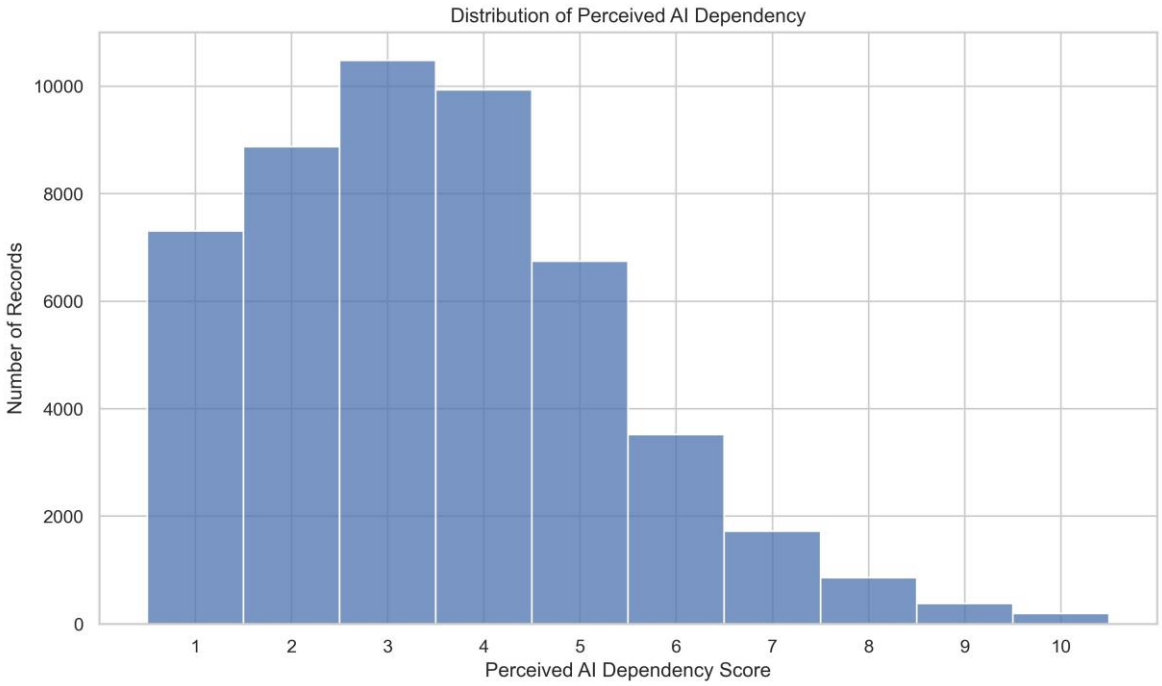


Figure 11.2: Distribution of perceived AI dependency.

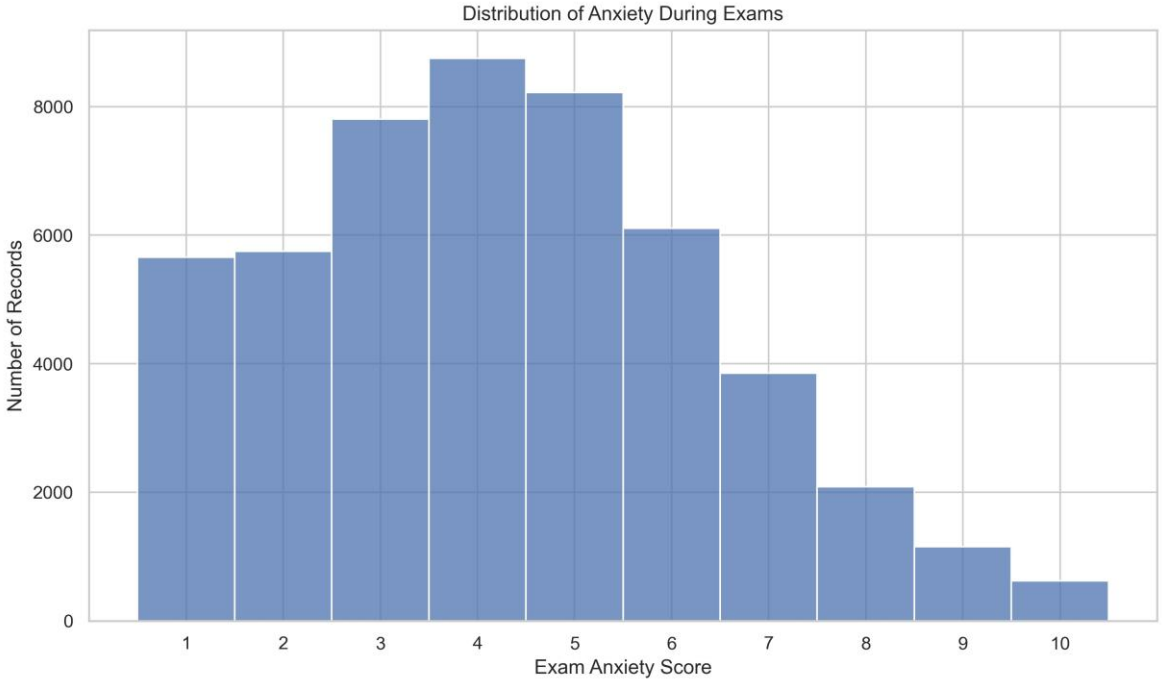


Figure 11.3: Distribution of exam anxiety levels.

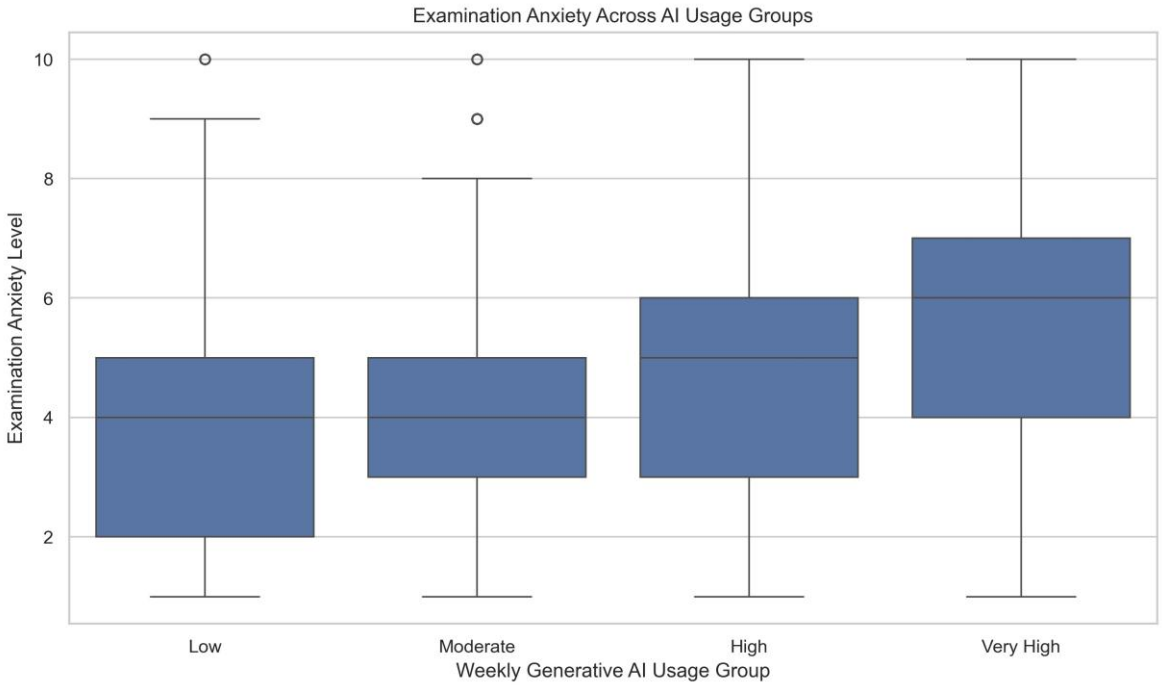


Figure 11.4: Exam anxiety across AI usage groups.

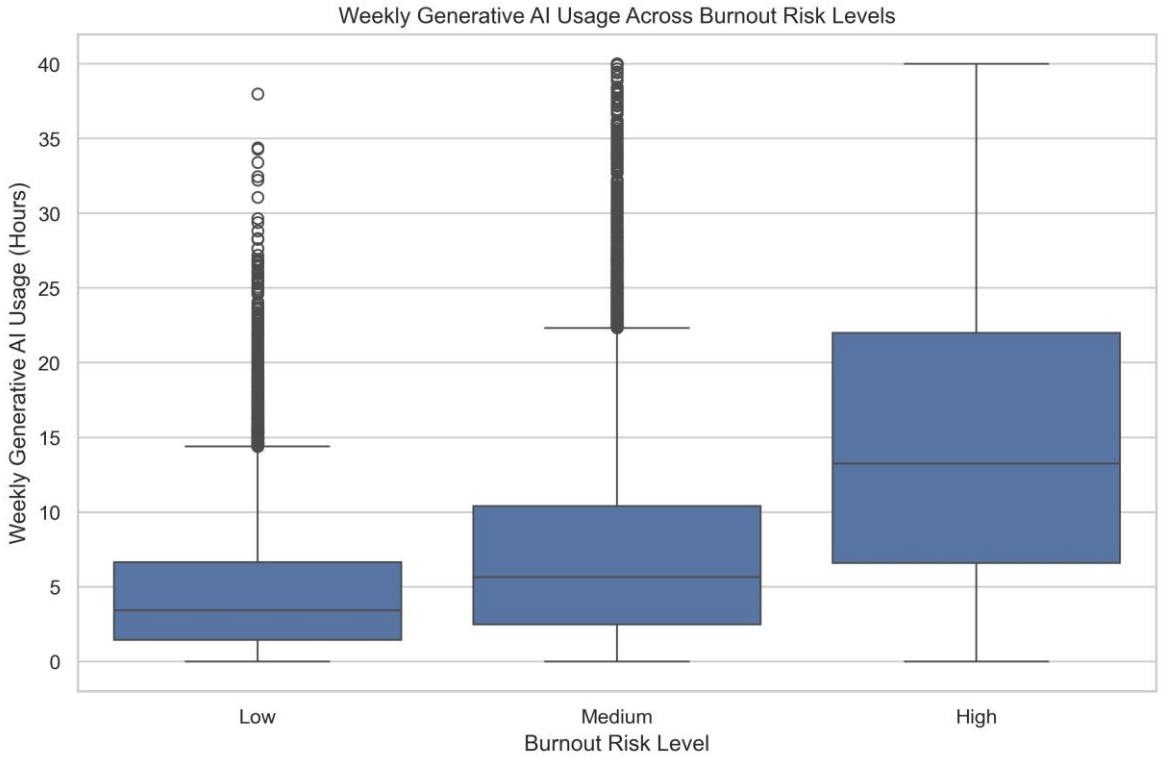


Figure 11.5: Weekly AI usage hours across burnout-risk categories.

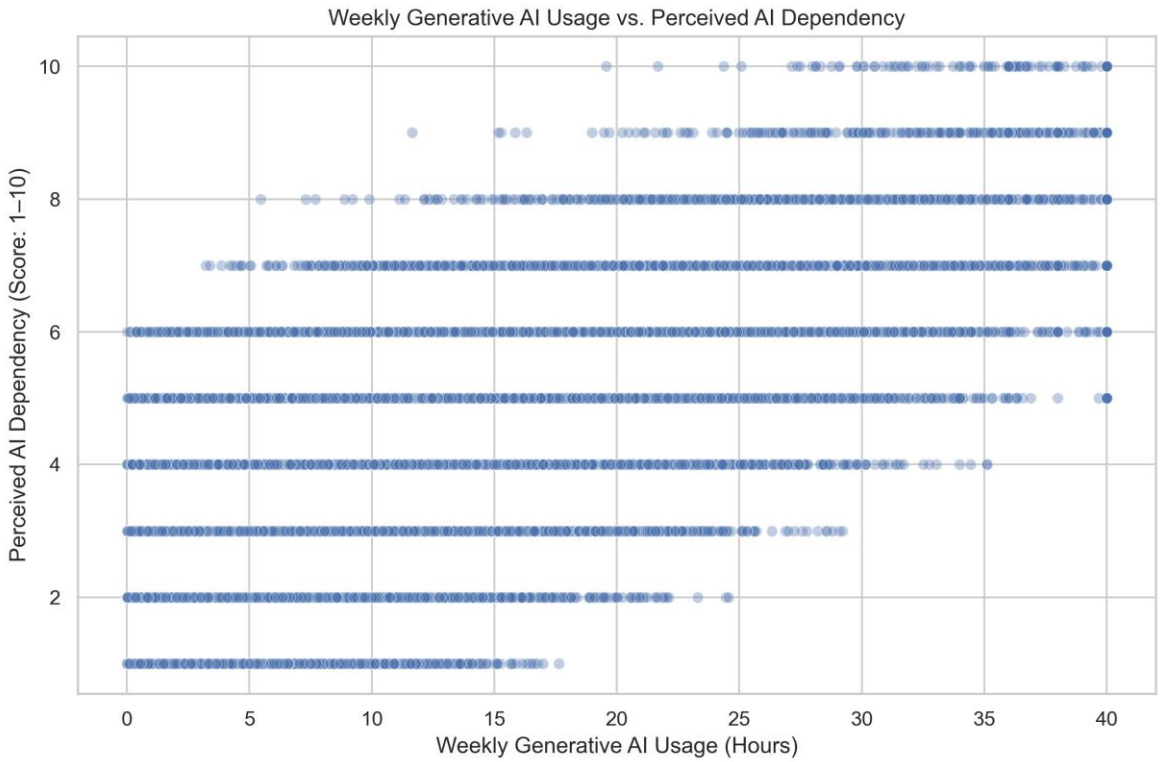


Figure 11.6: Relationship between AI usage hours and perceived dependency.

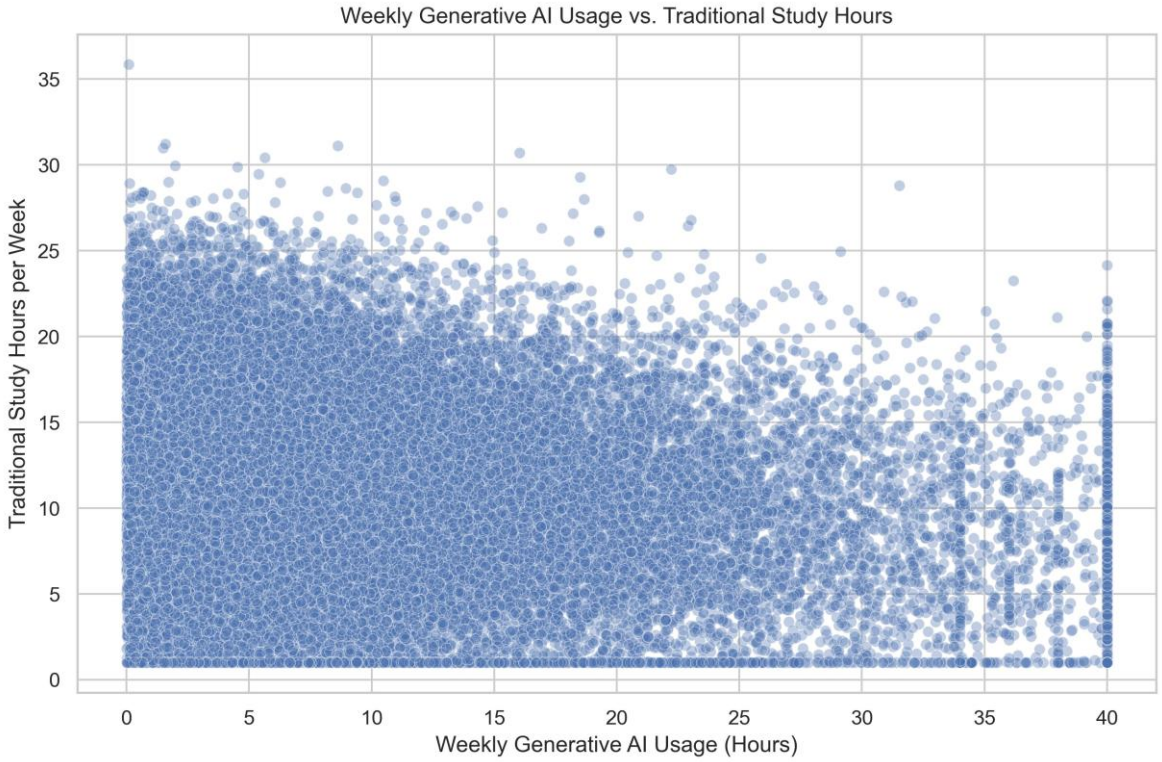


Figure 11.7: Relationship between AI usage and traditional study hours.

12. Skill Retention Analysis

Skill retention is examined across AI usage groups to identify differences in the distribution and typical values of the recorded scores. These comparisons do not establish whether AI usage improves or reduces skill retention.

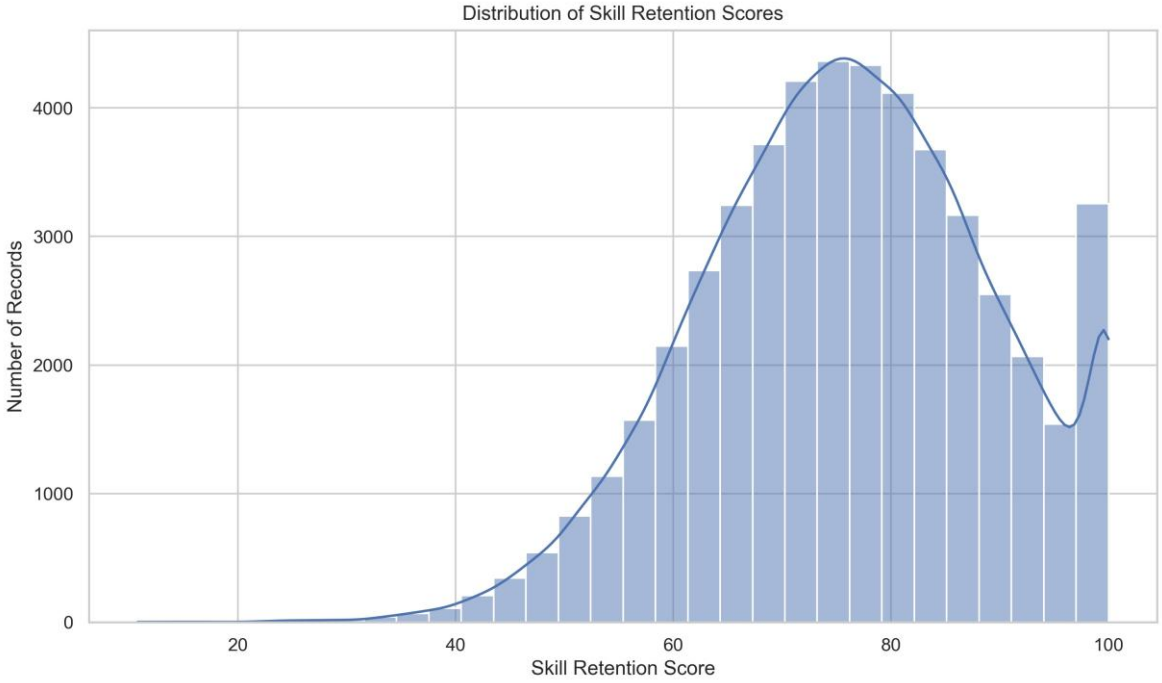


Figure 12.1: Distribution of skill-retention scores.

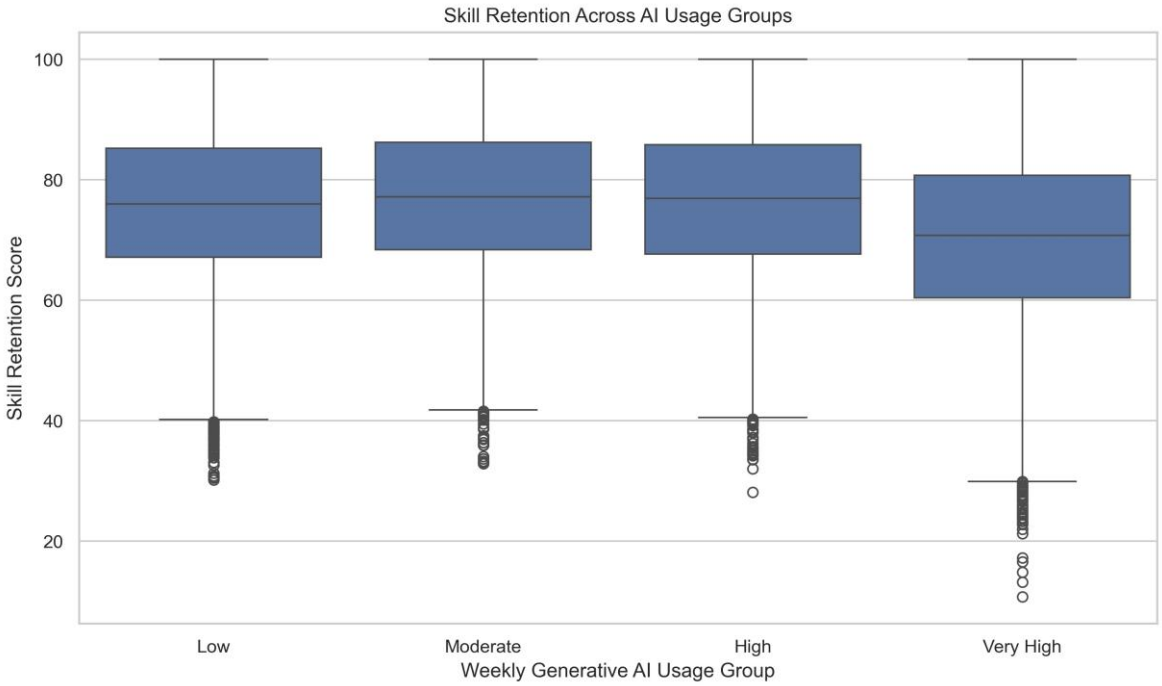


Figure 12.2: Skill-retention scores across AI usage groups.

13. Correlation Analysis

Variable pair	Pearson r
Pre-semester GPA vs. post-semester GPA	0.927
Weekly AI usage vs. perceived AI dependency	0.665
Perceived AI dependency vs. exam anxiety	0.308
Weekly AI usage vs. exam anxiety	0.269
Weekly AI usage vs. traditional study hours	-0.157
Weekly AI usage vs. post-semester GPA	-0.019

- Pearson's correlation coefficient measures the strength and direction of a linear association. A positive coefficient indicates that the variables tend to increase together; a negative coefficient indicates an inverse linear association. A coefficient near zero indicates little linear association, but does not rule out nonlinear relationships.
- Correlation does not imply causation. These exploratory coefficients do not, by themselves, establish statistical significance or control for confounding variables.



Figure 13.1 Correlation heatmap for numerical variables.

14. Additional Visualizations

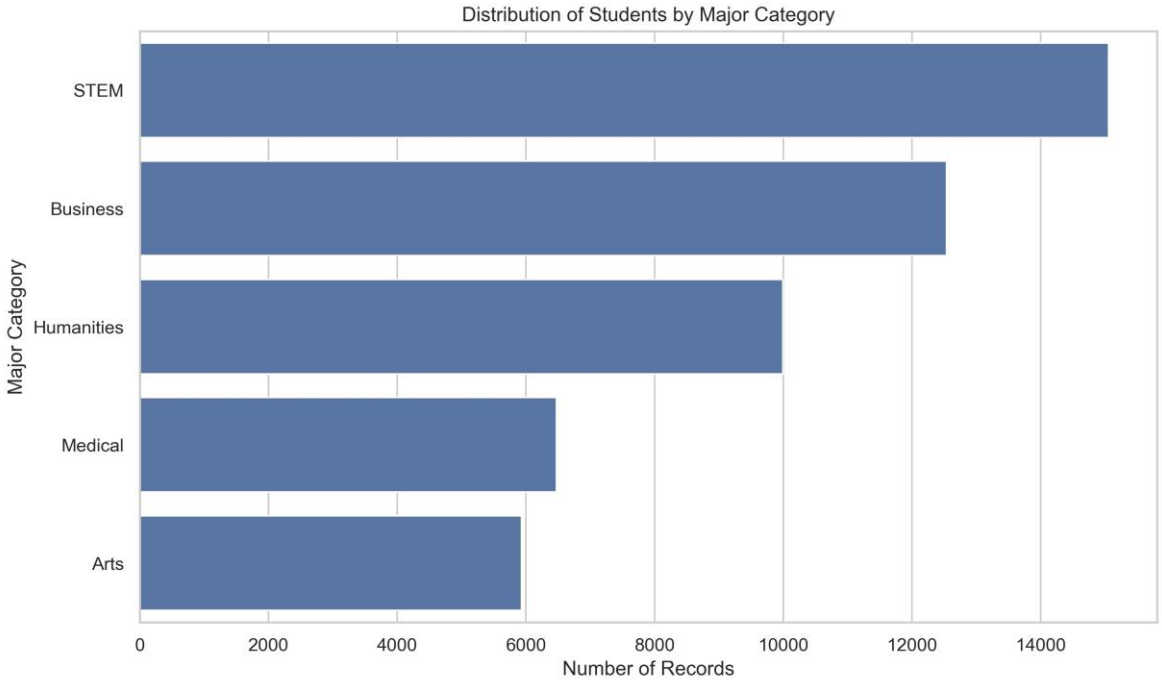


Figure 14.1: Distribution of students across major categories.

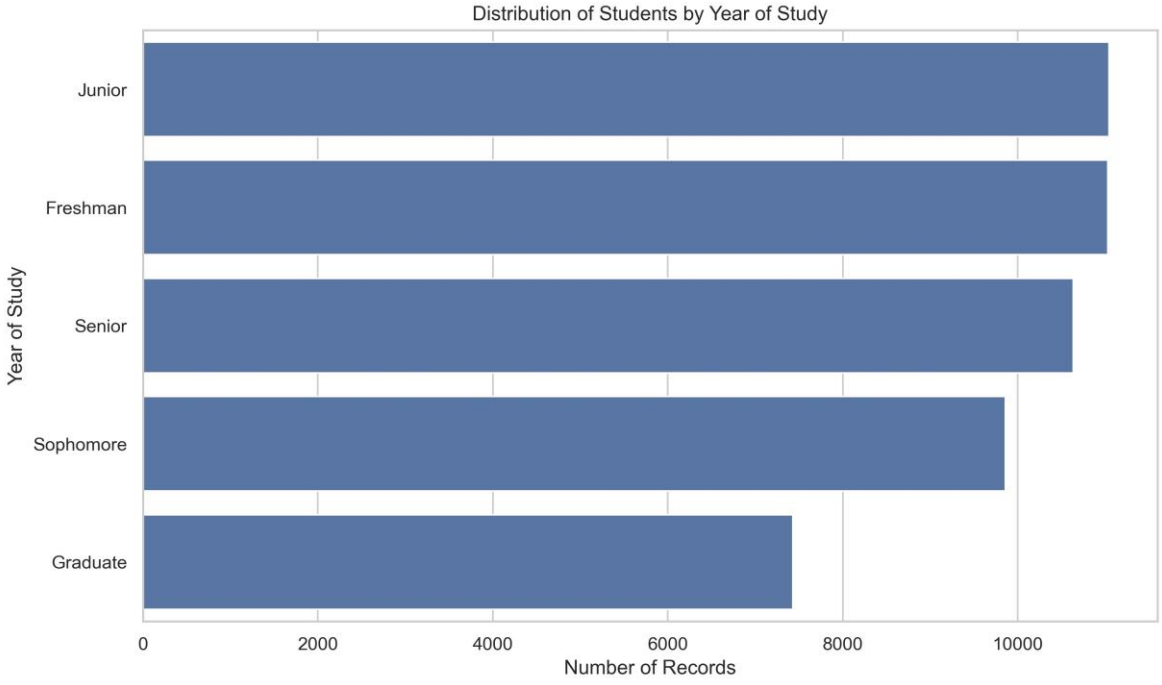


Figure 14.2: Distribution of students by year of study.

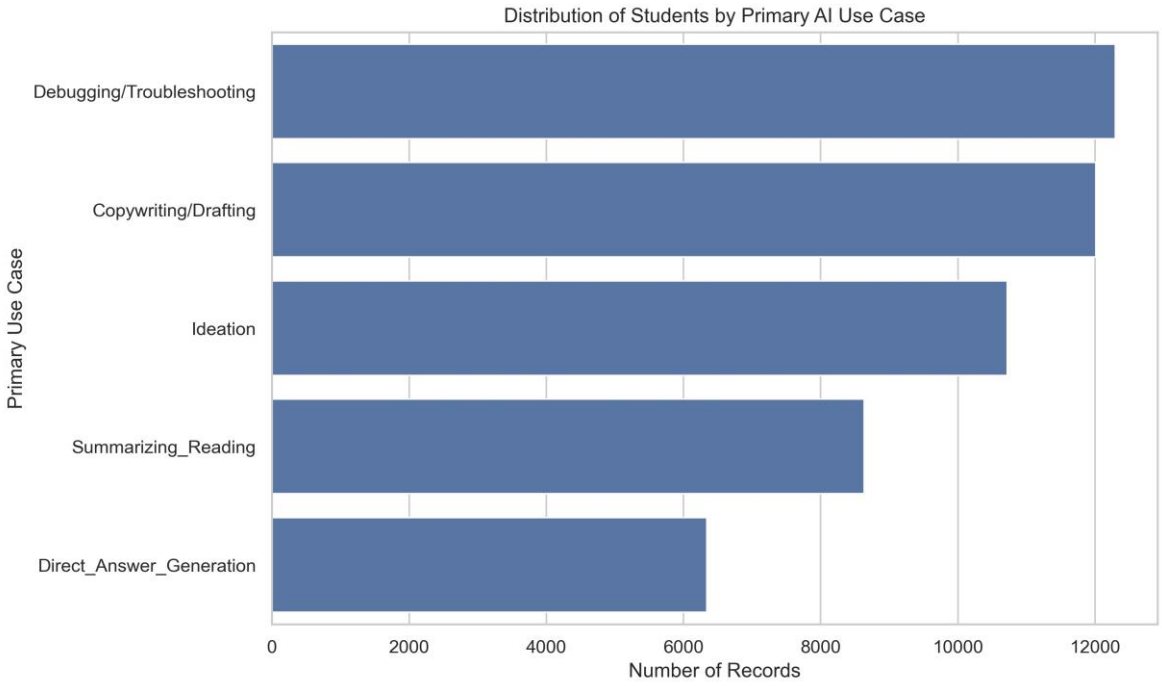


Figure 14.3: Distribution of primary generative AI use cases.

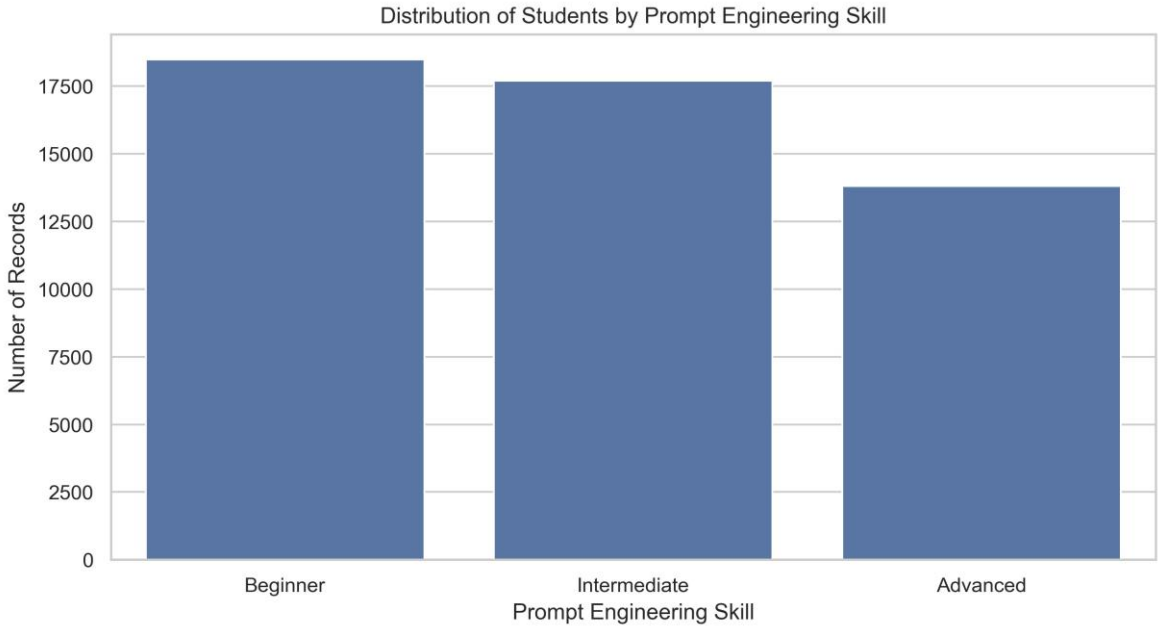


Figure 14.4: Distribution of prompt-engineering skill levels.

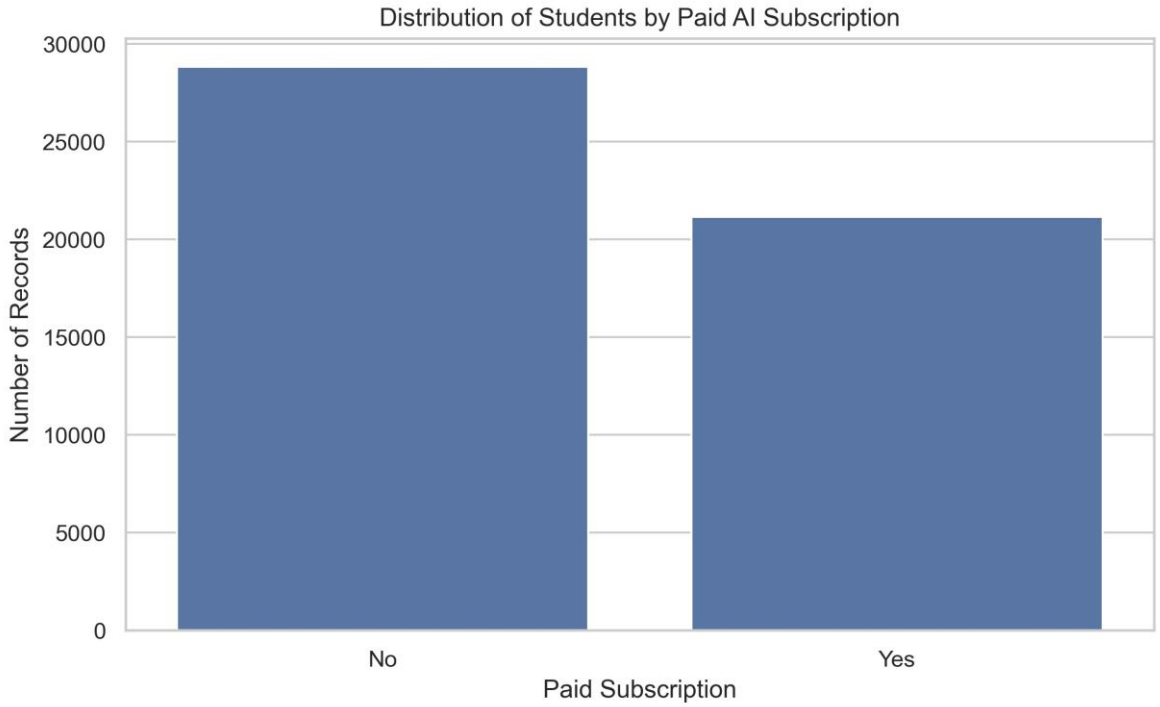


Figure 14.5: Distribution of paid subscription status.

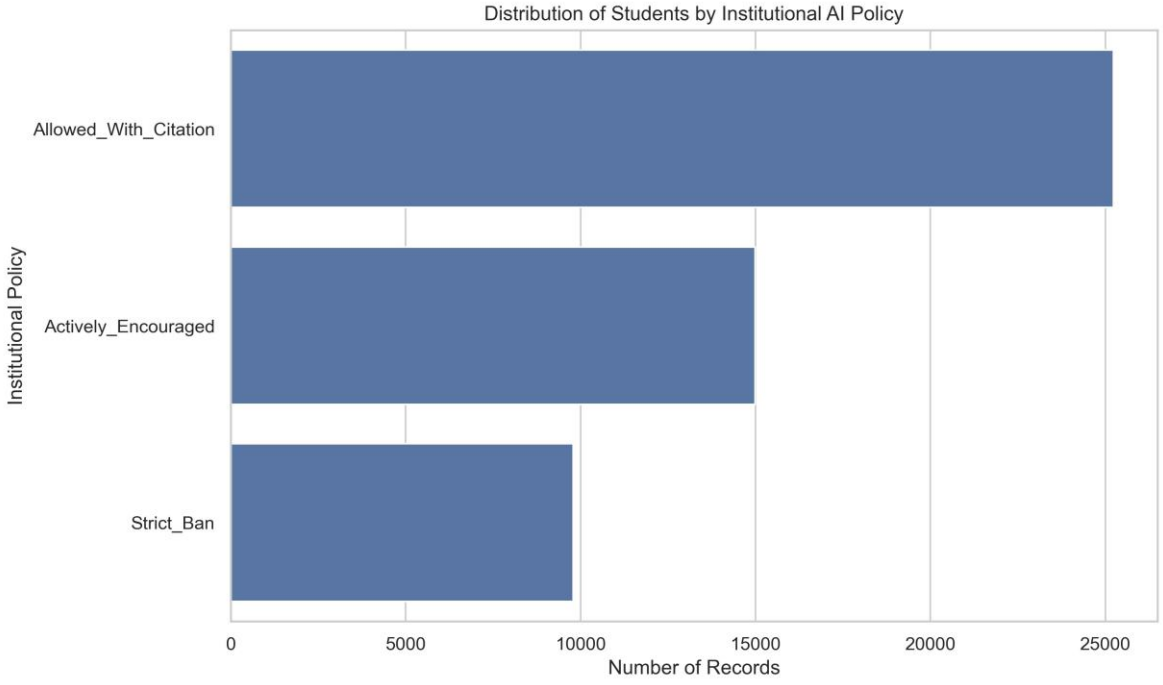


Figure 14.6: Distribution of institutional AI policies.

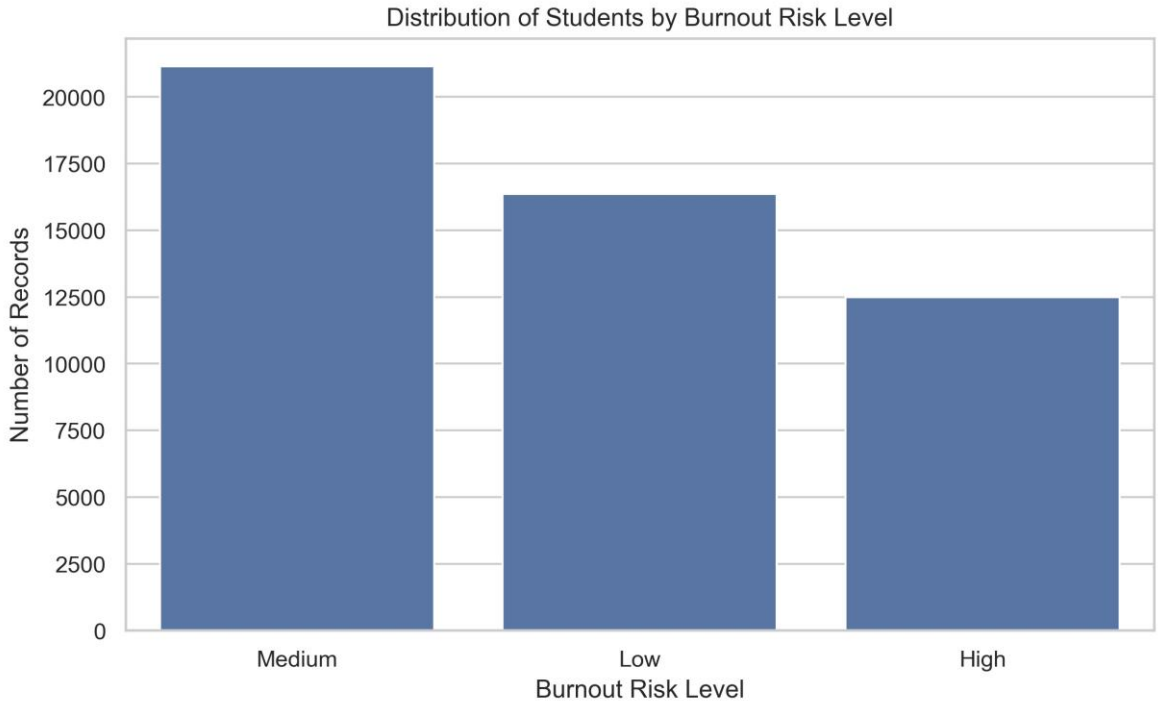


Figure 14.7: Distribution of burnout-risk categories.

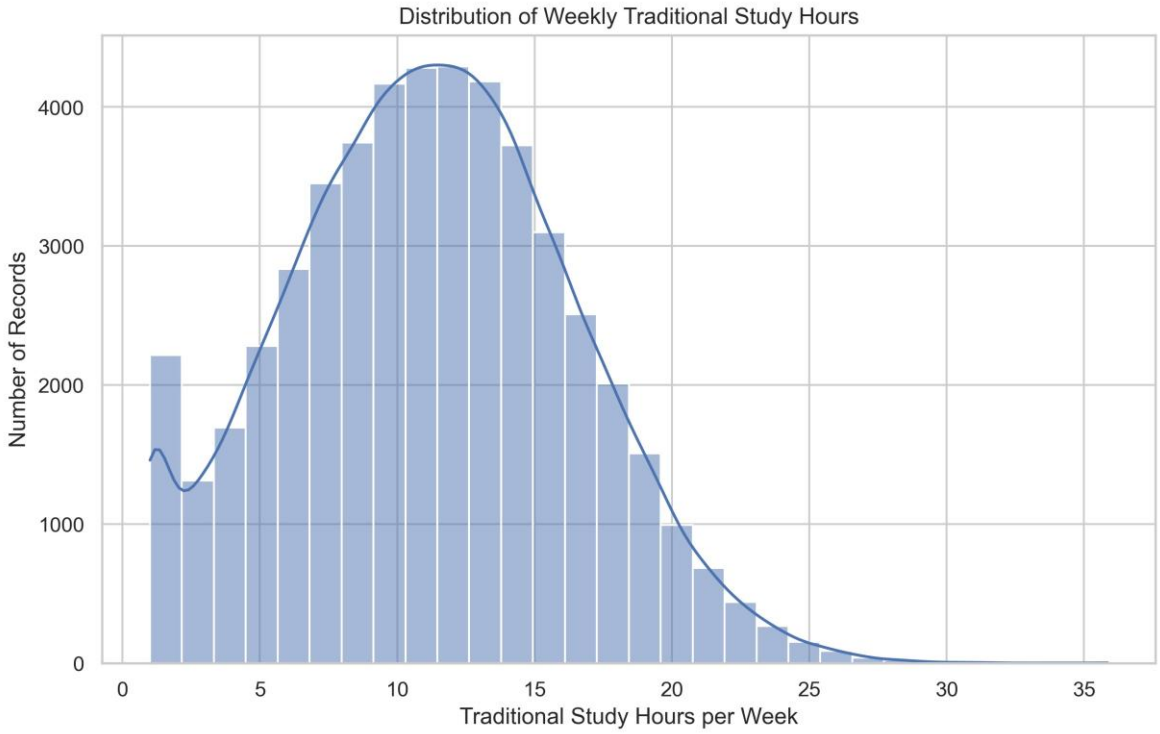


Figure 14.8: Distribution of traditional study hours.

15. Key Findings

- The processed dataset contains 50,000 records. The average pre-semester GPA is 3.15, compared with 3.35 for post-semester GPA.
- Among 50,000 records with valid paired GPA values, 43,759 (87.52%) show an increase, 49 (0.10%) are unchanged, and 6,192 (12.38%) show a decrease.
- The AI usage group counts are: Low: 22,592 (45.18%); Moderate: 12,173 (24.35%); High: 10,229 (20.46%); Very High:5,006 (10.01%).
- The Pearson correlation between weekly AI usage and perceived AI dependency is 0.665. This describes a linear association in the dataset and does not establish causality.
- The Pearson correlation between weekly AI usage and exam anxiety is 0.269. This result should be interpreted alongside the variable definitions, data quality, and possible confounding factors.
- The visualizations and group comparisons provide a descriptive overview of AI usage patterns, academic measures, and student well-being indicators. Findings should be verified against the dataset documentation before making broader claims.

16. Limitations

- The representativeness of the dataset cannot be established without documentation of its source and sampling method.
- Associations observed in this analysis do not establish causation.
- Potential confounding factors may affect academic performance, study habits, anxiety, and burnout risk.
- Pearson correlation captures linear associations and may not fully describe nonlinear relationships.
- The definitions and measurement methods for GPA, dependency, anxiety, skill retention, and burnout risk should beverified.
- The method used to derive AI_Usage_Group should be documented and checked against the raw weekly AI usagevalues.
- No predictive model or causal inference analysis is performed in this report.

17. Conclusion

This project applies Exploratory Data Analysis to investigate patterns in generative AI usage and student-related outcomes. It combines data-quality checks, descriptive statistics, categorical summaries, GPA comparisons, correlation analysis, and visualizations to provide an overview of the processed data.

The analysis provides a foundation for more detailed work, including hypothesis testing, subgroup comparisons, regression analysis, and predictive modeling. Any further analysis should begin by verifying the source, definitions, and reliability of the dataset. Conclusions should remain proportional to the evidence and should not treat observational associations as proof of cause and effect.

18. Tools and Methodology

Component	Purpose
Python	Data processing and analysis
CSV module	Loading the processed dataset
statistics	Descriptive statistics
ReportLab And Word	PDF report editing and generation
Matplotlib / Seaborn	Source of exported visualizations
Jupyter Notebook	Interactive EDA workflow